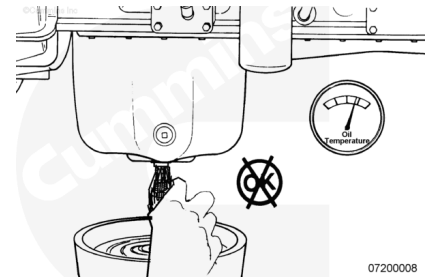


Trainings ▾

Drain

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.

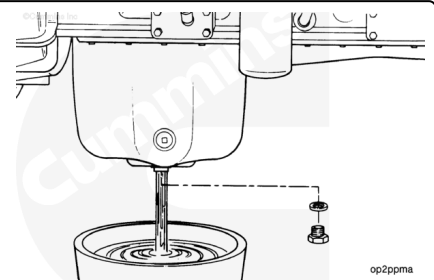


⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Operate the engine until the water temperature reaches 60°C [140°F]. Shut off the engine. Remove the oil drain plug from the bottom of the lubricating oil pan. Do **not** remove the plugs on either side of the oil pan to drain the oil. They will **not** allow the oil to drain completely.

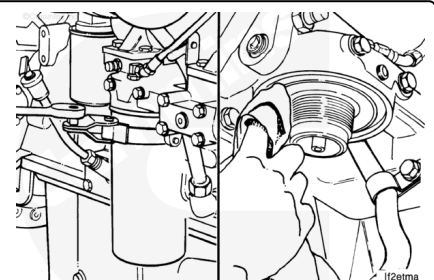
Note : Do not remove the plugs on either side of the oil pan to drain the oil. They will **not** allow the oil to drain completely.



Clean the area around the lubricating oil filter head.

Use oil filter wrench, Part Number 3375049, to remove the oil filter(s).

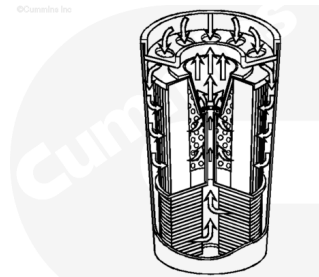
Clean the gasket surface of the oil filter head. The o-ring can stick on the filter head; **make sure** it is removed.



Fill

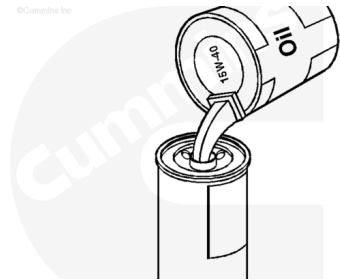
Use the correct oil filter for your engine. Refer to Procedure 102-002 in Section 2.

(/qs3/pubsys2/xml/en/procedures/35/35-102-002-om-mar.html)



⚠ CAUTION ⚠

If the filter canister is damaged in any way, do not use it. Dents or scrapes can lead to a rupture or premature failure of the filter.



⚠ CAUTION ⚠

Lack of lubrication during start-up is harmful to the engine.

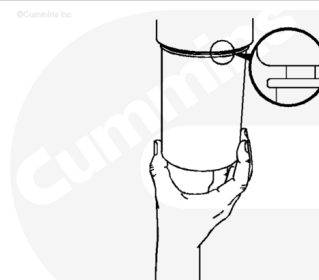
Fill the new oil filter(s) with clean engine oil.

⚠ CAUTION ⚠

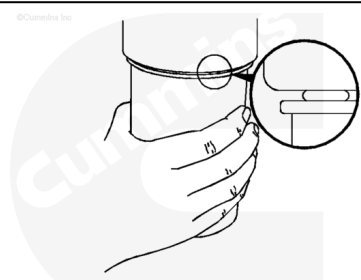
Mechanical overtightening of the filter can distort the threads or damage the filter element seal.

Install the new oil filter on the filter head.

Turn the filter until the gasket contacts the filter head surface.



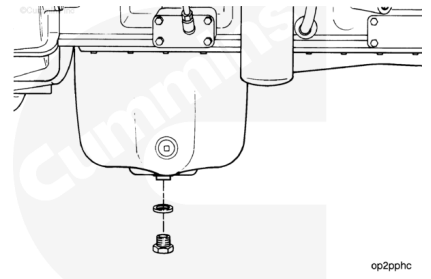
Tighten the filter an additional 1/2 to 3/4 of a turn after the gasket contacts the filter head surface, or as specified by the filter manufacturer.



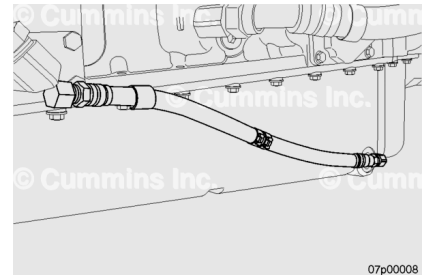
Clean and check the oil drain plug threads and the seal surface.

Install the oil drain plug in the lubricating oil pan.

Torque Value: 88 n•m [65 ft-lb]



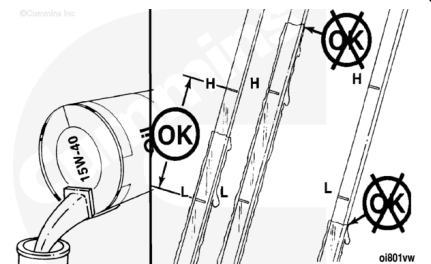
If a remote drain hose has been provided, and a change oil pump used, disconnect the hose from the pump and install the cap onto the end of the hose and secure the hose to the side of the lubricating oil pan using the clips that are provided.



Use a high-quality multiviscosity oil such as Valvoline® Premium Blue™, Valvoline® Premium Blue™ 2000, or equivalent. Choose the correct oil for a specific operating climate. Refer to Procedure 018-003 Section V.
(/qs3/pubsys2/xml/en/procedures/101/101-018-003-om-mar.html)

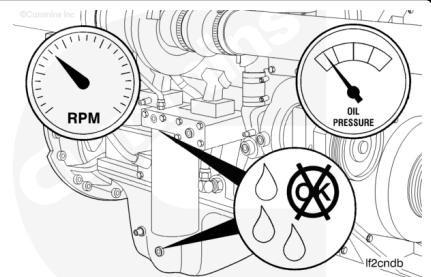


Fill the engine with clean oil to the H (high) mark on the dipstick. For total system capacity including filter, Refer to Procedure 018-017 in Section V.
(/qs3/pubsys2/xml/en/procedures/35/35-018-017-om-mar.html)



⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level is in the oil pan.



Operate the engine at idle speed to inspect for leaks at the filter and the drain plugs.

Shut off the engine. Wait approximately 10 minutes for the oil to drain back from the upper parts of the engine to the oil pan.

Check the oil level again. Add oil as necessary to bring the level up to the "H" (high) mark on the dipstick.

